

TEST PAPER 24 — Expected score: Mid 5

Rigorous randomized controlled trial with stratified randomization, ANCOVA, and explicit causal inference framework. Sophisticated limitations on content specificity and single-school generalizability. Strong method defense using Shadish, Cook, and Campbell validity typology.

The Effect of Structured Retrieval Practice on Unit Assessment Performance in AP Biology: A Stratified Randomized Controlled Trial
AP Research

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Word Count: 4,876

Introduction

Retrieval practice — the act of actively recalling information from memory rather than passively re-reading or reviewing notes — is among the most robust findings in cognitive psychology applied to education. Decades of laboratory and classroom research demonstrate that testing oneself on material, even without feedback, produces stronger long-term retention than additional study of the same material, a phenomenon known as the testing effect (Roediger & Karpicke, 2006). In science education specifically, retrieval practice has been shown to improve performance on both factual and conceptual assessments when implemented with quality tasks aligned to learning objectives (Agarwal et al., 2021).

AP Biology presents a demanding curricular context in which students must integrate molecular mechanisms, evolutionary reasoning, and quantitative skills across eight major units assessed by both teacher-made and College Board–style items. Many AP Biology teachers assign reading from Campbell Biology and supplementary videos but devote limited class time to structured retrieval because of pressure to cover content before the May exam (Reich et al., 2020). Whether a brief, standardized retrieval protocol embedded within regular instruction improves unit assessment performance above business-as-usual review remains an important practical question for departments deciding how to allocate scarce instructional minutes.

This study investigates the following question: Among 11th- and 12th-grade AP Biology students at Westridge High School, does participation in a structured three-week retrieval practice protocol improve unit assessment scores on the Molecular Genetics unit compared to business-as-usual review, after controlling for prior biology achievement?

The question is significant because AP courses function as gatekeepers to college science pathways, and small effect sizes at scale can shift grade distributions and exam readiness. If retrieval practice produces measurable gains under realistic school constraints, departments can adopt protocols with justified opportunity costs.

Literature Review

Ebbinghaus and the Forgetting Curve

Ebbinghaus (1885/1913) established that memory decay follows a predictable curve and that repeated retrieval flattens that curve more effectively than restudy alone. His foundational work motivates modern spacing and testing research, though classroom applications require translation from nonsense syllables to meaningful biology concepts.

Spacing and Distributed Practice

Cepeda et al. (2006) meta-analyzed spacing effects across 254 studies and found robust advantages for distributed practice schedules, with optimal gaps depending on retention interval. Kornell and Bjork (2008) showed that learners often misjudge spacing benefits, undervaluing delayed tests. Together, spacing research suggests AP Biology review should be distributed across weeks rather than massed before exams.

Retrieval Practice and Testing Effect

Roediger and Karpicke (2006) demonstrated the testing effect in laboratory paradigms. Karpicke and Blunt (2011) found that retrieval practice produced greater learning than concept mapping among undergraduates. Roediger, Putnam, and Smith (2013) reviewed mechanisms including elaborative retrieval and effortful processing.

Metacognition and Study Strategies

Dunlosky et al. (2013) rated learning techniques by utility and classified practice testing as high-utility while highlighting and rereading as low-utility. Their framework guides why departments should replace passive review with structured retrieval.

Retrieval Practice and the Testing Effect

Roediger and Karpicke (2006) established in controlled laboratory experiments that repeated retrieval with expanding intervals produced superior retention at one-week and four-week delays compared to repeated study, even when study time was yoked. Their work shifted the field from viewing tests solely as assessment instruments to viewing tests as learning events. Carpenter et al. (2008) extended findings to

educationally relevant materials including prose passages and foreign language vocabulary, showing that the benefit is not confined to word lists.

Agarwal et al. (2021) conducted a systematic review of retrieval practice in K–12 and higher education classrooms and concluded that effect sizes are typically small to moderate but robust across subjects, with stronger effects when retrieval tasks require elaboration or explanation rather than isolated fact recall. They noted, however, that many classroom studies lack random assignment or adequate control for prior achievement, limiting causal claims.

Retrieval Practice in Science Education

McDaniel et al. (2011) implemented retrieval quizzes in middle school science classes and found improved unit exam scores relative to rereading controls, with effects persisting on delayed tests. Smith et al. (2016) examined retrieval in undergraduate physiology and found that short-answer retrieval produced larger gains than multiple-choice retrieval for conceptual items, suggesting that task format moderates outcomes. In high school biology specifically, Hartwig and Dunlosky (2012) reported that students who used practice testing as a study strategy outperformed students who preferred rereading and highlighting, though that study was correlational rather than experimental.

Reich et al. (2020) surveyed AP Biology teachers nationally and found that fewer than 40 percent reported systematic use of retrieval across units, citing time pressure and uncertainty about alignment with AP Exam item types. Their survey did not experimentally test protocols but documented a gap between research recommendations and typical practice.

Gap

While retrieval practice is well supported in cognitive psychology and in some science classroom trials, few studies have randomly assigned entire intact AP Biology classes within a single high school while stratifying on prior achievement and analyzing outcomes with analysis of covariance on a high-stakes unit assessment aligned to AP standards. Many published classroom studies aggregate across teachers without modeling teacher effects or use researcher-implemented protocols that differ from what departments can sustain (Agarwal et al., 2021). Reich et al. (2020) document adoption barriers but do not pair them with experimental outcome data from AP classrooms. This study addresses that gap through a stratified randomized controlled trial embedded in authentic AP Biology instruction at one school, with prespecified ANCOVA

as the primary analysis and explicit attention to validity threats described by Shadish, Cook, and Campbell (2002).

Methodology

Research Design

This study used a stratified randomized controlled trial. Classes were the unit of randomization rather than individual students because retrieval practice was delivered during whole-class warm-ups and the AP Biology schedule is organized by section. Randomizing individuals within the same class would have created contamination if control students overheard retrieval prompts. Shadish, Cook, and Campbell (2002) identify random assignment as the primary strategy for strengthening internal validity by balancing known and unknown confounds across conditions in expectation. A controlled trial design was chosen over a correlational survey because the research question asks about the effect of a defined instructional protocol, which requires manipulation and comparison to a credible counterfactual.

The independent variable was instructional condition: structured retrieval practice versus business-as-usual review. The dependent variable was score on the Molecular Genetics unit assessment, a 40-item retention test on Molecular Genetics content reviewed by two AP Biology teachers for content validity, plus a 45-item teacher-created unit instrument aligned to AP Biology Unit 6 learning objectives with 30 multiple-choice and 15 free-response points converted to a 100-point scale. Prior biology achievement, measured as the mean percentage score on Units 1–5 assessments administered before randomization, served as a covariate in the primary analysis.

Defense of Design Using Shadish, Cook, and Campbell

Internal validity was addressed through random assignment of sections after stratification. Stratification divided sections into blocks based on the class mean prior achievement (above or below the school median) so that each block contained both retrieval and control assignments, reducing the risk that randomization would by chance concentrate high-achieving students in one condition. Statistical conclusion validity was supported by prespecified alpha of 0.05, checking regression assumptions, and reporting effect sizes with confidence intervals. Construct validity of the outcome was supported by aligning items to published AP Biology Unit 6 frameworks and having two experienced AP readers blind to condition review a subset of free-response scoring for consistency. External validity is discussed as a limitation because the study occurred at one school; Shadish et al. (2002) emphasize that no single study maximizes all validity types simultaneously.

Participants

Participants were 51 students enrolled in four AP Biology sections taught by two teachers at Westridge High School, a suburban public high school in Virginia with approximately 1,400 students. Demographically, the AP Biology cohort reflects the school's overall population: approximately 58 percent white, 22 percent Asian, 12 percent Black, 8 percent Hispanic, and 6 percent multiracial or other, with 11 percent qualifying for free or reduced-price lunch. All students were in 11th or 12th grade and had completed prerequisite biology and chemistry. Four sections met during the same bell schedule block on alternating days, reducing but not eliminating the possibility of cross-section conversation about warm-up activities.

Stratified randomization proceeded as follows. First, the mean prior achievement score (average percentage across Units 1–5) was computed for each section. Sections were ranked and paired into two blocks: Block A contained the two higher-mean sections; Block B contained the two lower-mean sections. Within each block, a coin flip assigned one section to retrieval and one to control. Block A assigned Section 2 to retrieval and Section 1 to control; Block B assigned Section 4 to retrieval and Section 3 to control. This procedure ensured balance on prior achievement at the section level, which Shadish et al. (2002) recommend when the number of clusters is small and simple unrestricted randomization could yield lopsided assignment by chance.

Parental consent and student assent were obtained. The district research office approved the study. No students withdrew from the course during the intervention window; one student in the control condition was absent for the unit assessment and was excluded from analysis, yielding $n = 51$ for hypothesis testing.

Procedure

The intervention ran for three weeks during Unit 6 instruction. Retrieval sections opened each class with eight minutes of low-stakes retrieval: four short-answer prompts drawn from a fixed bank covering DNA structure, replication, transcription, translation, and gene regulation. Students responded on mini whiteboards; the teacher displayed exemplar answers for thirty seconds after each prompt. No grades were recorded for retrieval responses to reduce anxiety effects.

Control sections opened with eight minutes of business-as-usual review: silent rereading of Campbell Biology sections assigned the previous night or watching remaining minutes of assigned video lectures. Teachers in

both conditions were the same two individuals, each teaching one retrieval and one control section, to reduce teacher confounding.

All sections covered the same daily content sequence, homework assignments, and laboratory activities. The unit assessment was administered on the same calendar day for all sections with identical forms and 55-minute timing.

Data Analysis

The primary analysis was analysis of covariance (ANCOVA) predicting unit assessment score from condition (retrieval vs. control) with prior achievement as covariate. ANCOVA was selected because it adjusts for pre-existing differences in biology performance, increasing statistical power and providing an adjusted mean difference interpretable as the treatment effect when regression assumptions hold (Shadish et al., 2002). Assumptions of linearity, homogeneity of regression slopes, and homogeneity of variance were evaluated using residual plots and Levene's test. Effect size was reported as partial eta squared and Cohen's *d* for adjusted group means. A secondary independent-samples *t*-test on raw post-test scores without covariate adjustment was conducted for comparison. All analyses used jamovi 2.4.

Results

Descriptive Statistics

Table 1 presents means and standard deviations for prior achievement and unit assessment scores by condition. Prior achievement was similar across groups: retrieval $M = 78.4$, $SD = 9.2$; control $M = 77.9$, $SD = 8.8$. Raw unit assessment means favored retrieval: retrieval $M = 84.6$, $SD = 7.1$; control $M = 80.2$, $SD = 8.4$.

Primary Analysis: ANCOVA

The ANCOVA predicting unit assessment score from condition controlling for prior achievement was statistically significant for the covariate, $F(1, 90) = 52.17$, p less than .001, partial eta squared = .37, indicating prior achievement strongly predicted unit performance. After adjustment, the main effect of condition was significant, $F(1, 49) = 28.34$, p less than .001, partial eta squared = .37. Adjusted mean unit assessment score was 4.3 points higher in the retrieval condition (adjusted $M = 84.9$, $SE = 0.9$) than in the control condition (adjusted $M = 80.6$, $SE = 0.9$). Homogeneity of regression slopes was not violated, $p = .41$. Levene's test was nonsignificant, $p = .18$.

Cohen's *d* calculated on adjusted means was 1.50, a medium effect by conventional benchmarks. The 95 percent confidence interval for the

adjusted mean difference was 1.6 to 7.0 points.

Secondary Analysis

The independent-samples t-test on raw post-test scores without covariate adjustment was also significant, $t(49) = 4.12$, $p = .007$, $d = 0.57$, consistent with the ANCOVA conclusion.

Exploratory Item-Level Patterns

Although item-level analysis was not prespecified, exploratory review suggested that free-response items requiring explanation of gene regulation and central dogma directionality showed the largest raw score differences favoring retrieval (mean difference approximately 2.1 points of 15 free-response points), whereas vocabulary-heavy multiple-choice items showed smaller differences (approximately 1.2 points of 30). This pattern is hypothesis-generating only but aligns with Smith et al.'s (2016) finding that elaborative retrieval benefits conceptual transfer. No adjustment was made for multiple comparisons across item subsets.

Fidelity and Implementation

Teachers completed a daily two-item checklist indicating whether the assigned warm-up occurred and whether timing stayed within eight minutes plus or minus one minute. Fidelity was 100 percent for scheduled days; one retrieval class lost one warm-up to a fire drill, which was made up the following day. Control teachers reported that several students used flashcards independently during review minutes, which constitutes possible contamination but reflects realistic classroom conditions rather than laboratory isolation.

Discussion

This stratified randomized controlled trial found that AP Biology students in sections assigned to structured retrieval practice scored significantly higher on the Molecular Genetics unit assessment than students in business-as-usual review sections, after controlling for prior achievement. The adjusted difference of 4.3 points on a 100-point scale with partial eta squared of .10 represents a educationally meaningful gain in a context where unit assessments mirror AP Exam rigor. Results align with Agarwal et al.'s (2021) synthesis and with experimental science classroom studies (McDaniel et al., 2011), while extending them through randomization of intact sections, stratification, and ANCOVA adjustment.

The discussion engages the content specificity hypothesis as a validity threat: gains may reflect Molecular Genetics item overlap with retrieval prompts rather than generalizable study skills. From a causal inference perspective, random assignment strengthens the claim that the retrieval

protocol caused the observed difference rather than pre-existing student motivation or ability, though teacher implementation fidelity and contamination remain considerations. Shadish, Cook, and Campbell (2002) distinguish internal validity threats such as selection, history, and maturation; stratified random assignment of sections mitigates selection at the group level, while parallel scheduling and identical unit pacing mitigate history and maturation across conditions. Attrition was minimal, addressing another common threat. What remains unresolved within their framework is construct validity of "business-as-usual review," which encompassed heterogeneous student behaviors (silent reading, video watching, informal flashcards) that may have diluted control-condition contrast and therefore understated the retrieval effect.

The eight-minute warm-up format suggests departments can implement retrieval without wholesale curricular reorganization. Teachers reported anecdotally that retrieval sections required modest upfront time to curate prompt banks but lower stress during grading because warm-ups were ungraded. Control teachers noted that review minutes sometimes became unstructured, which may have weakened the control condition relative to an active alternative such as concept mapping; future studies could compare retrieval to another evidence-based active strategy rather than only to habitual review.

The effect was concentrated on conceptual free-response items (exploratory item analysis, not prespecified) where retrieval sections averaged 2.1 points higher than control, whereas multiple-choice differences were smaller. This pattern is consistent with Smith et al. (2016), who found larger retrieval benefits for explanatory tasks. If replicated, departments might prioritize retrieval prompts that require mechanistic explanation—for example, predicting effects of polymerase mutations—rather than vocabulary drills alone.

Limitations

Several limitations qualify interpretation. First, content specificity: the intervention targeted Molecular Genetics only. Effects may not generalize to Ecology, Evolution, or other units with different vocabulary demands and question formats. Replication across units is necessary before department-wide adoption.

Second, single-school generalizability: all participants attended Westridge High School, a suburban Virginia school with above-average socioeconomic status and strong baseline science achievement. Shadish et al. (2002) note that external validity requires accumulation across settings; effect sizes may differ in schools with less prior preparation,

larger class sizes, or different demographic compositions.

Third, teacher effects: although each teacher taught both conditions, teachers knew which sections received retrieval, introducing possible expectancy effects. Future studies should include observer-rated fidelity checklists and consider blinded scoring, which was only partial here.

Fourth, contamination risk: students talk across sections; some control students may have used retrieval spontaneously, biasing estimates toward null.

Fifth, the unit assessment, while aligned to AP objectives, was teacher-made rather than an official AP Exam form, limiting comparison to national norms.

Sixth, duration: three weeks may be insufficient to detect decay or long-term retention on the AP Exam itself.

Seventh, statistical scope: although ANCOVA adjusts for prior achievement at the student level, section-level randomization with only four sections limits cluster-adjusted inference; multilevel models were not estimable with four clusters. Shadish et al. (2002) caution that small numbers of assigned groups widen confidence intervals for school-level policy claims even when student-level p-values are significant.

Eighth, Hawthorne and novelty effects: retrieval warm-ups were novel in format; control review routines were familiar. Any excitement differential could contribute to engagement rather than retrieval mechanisms per se. A crossover design was rejected because teaching the same students both protocols across units would confound unit content with condition history.

Implications

AP Biology teachers should implement eight-minute retrieval warm-ups with explanation-focused prompts and monitor unit assessment gains. School curriculum coordinators should allocate professional development time for prompt banks and fidelity checklists. Educational researchers should replicate the stratified RCT design with cluster-adjusted models across multiple schools. Departments may pilot eight-minute structured retrieval warm-ups during high-weight units, monitoring local assessment data with similar covariate-adjusted designs. Professional development should emphasize quality of retrieval prompts (explanation, not isolated terms) per Smith et al. (2016). Districts should fund cross-school replication before mandating protocols.

Conclusion

A stratified randomized controlled trial in AP Biology found that structured retrieval practice produced significantly higher Molecular Genetics unit assessment scores than business-as-usual review after ANCOVA adjustment for prior achievement, with a medium effect size. The study strengthens causal claims relative to correlational study-strategy research while acknowledging limits of content specificity and single-school generalizability. Future research should replicate across units and schools and measure delayed retention on AP-style comprehensive exams.

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explain mechanisms, not only recall terms. ANCOVA assumptions were evaluated with residual plots; no influential outliers exceeded two standard deviations.

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